Fraud Detection in Financial Transactions

Data Science Case - Artefact | Updated results

- Leonardo Correia
- EDA, behavioral features, model selection and budget-constrained review
- 2025 history -> validated ranking -> January 2026 delivery

From fraud scores to an affordable review queue.

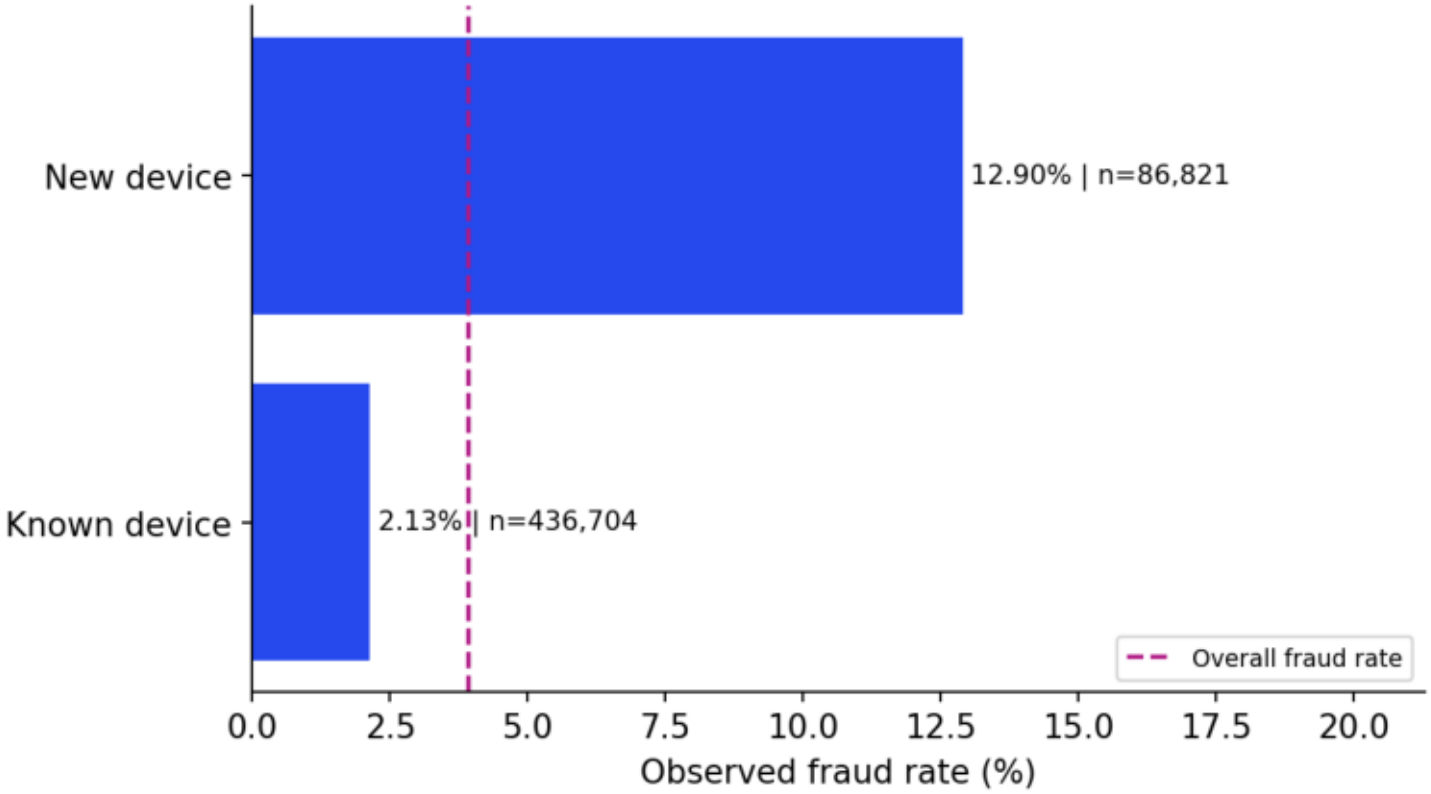
Objective and data overview

- 523,525 labeled transactions from 2025.
- 20,485 confirmed fraud cases: 3.91% prevalence.
- No missing values in the original historical columns.
- 44,451 January 2026 transactions require classification.
- Evaluate ranking quality and fraud recovery within review capacity.

Accuracy alone would hide the impact of the 3.91% fraud class.

EDA: device novelty is a strong risk signal

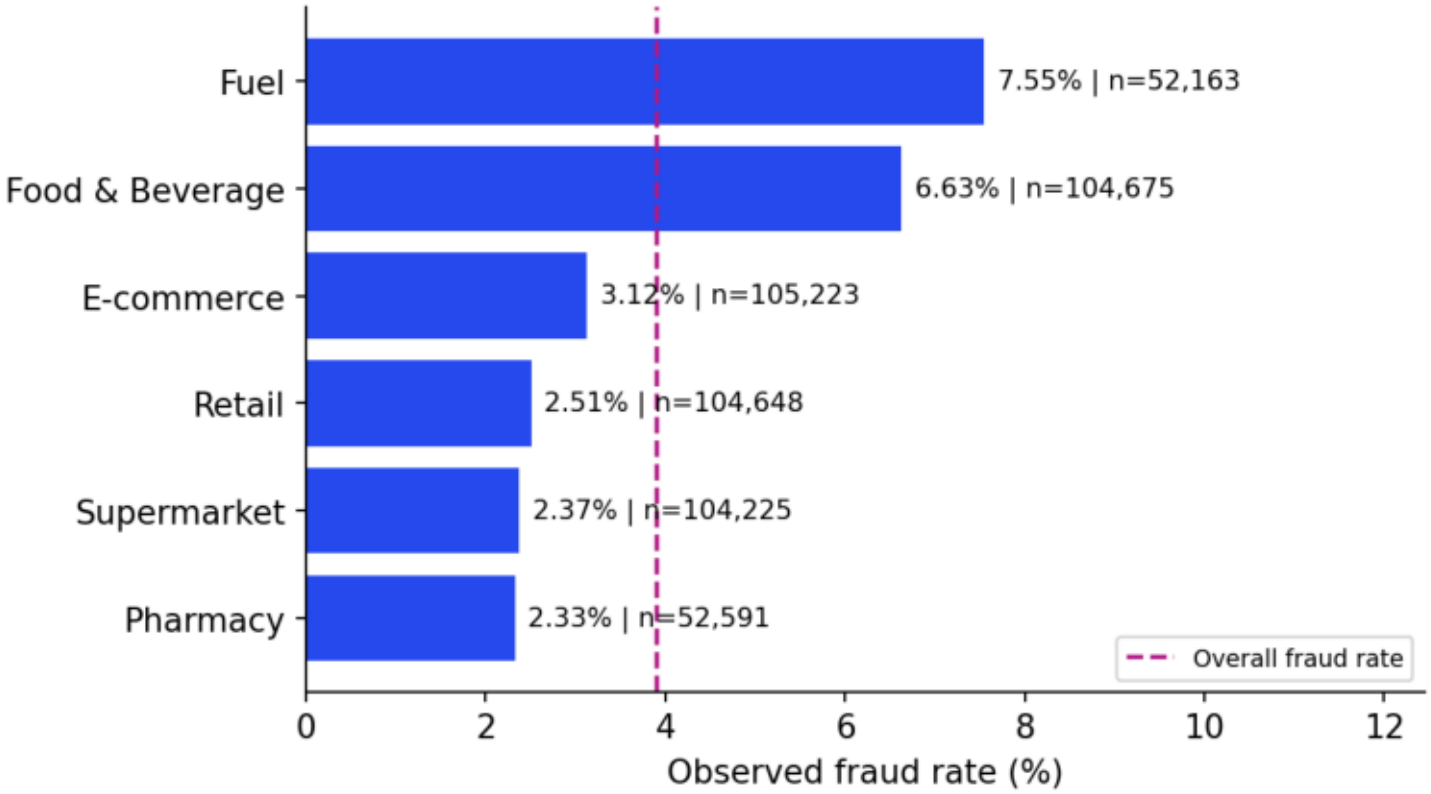
- New-device fraud rate: 12.90%.
- Known-device fraud rate: 2.13%.
- Approximately 6.1x higher observed risk for new devices.
- Association motivates a candidate feature; it does not prove causality.



Device novelty separates risk groups with substantial transaction support.

EDA: merchant type and timing matter

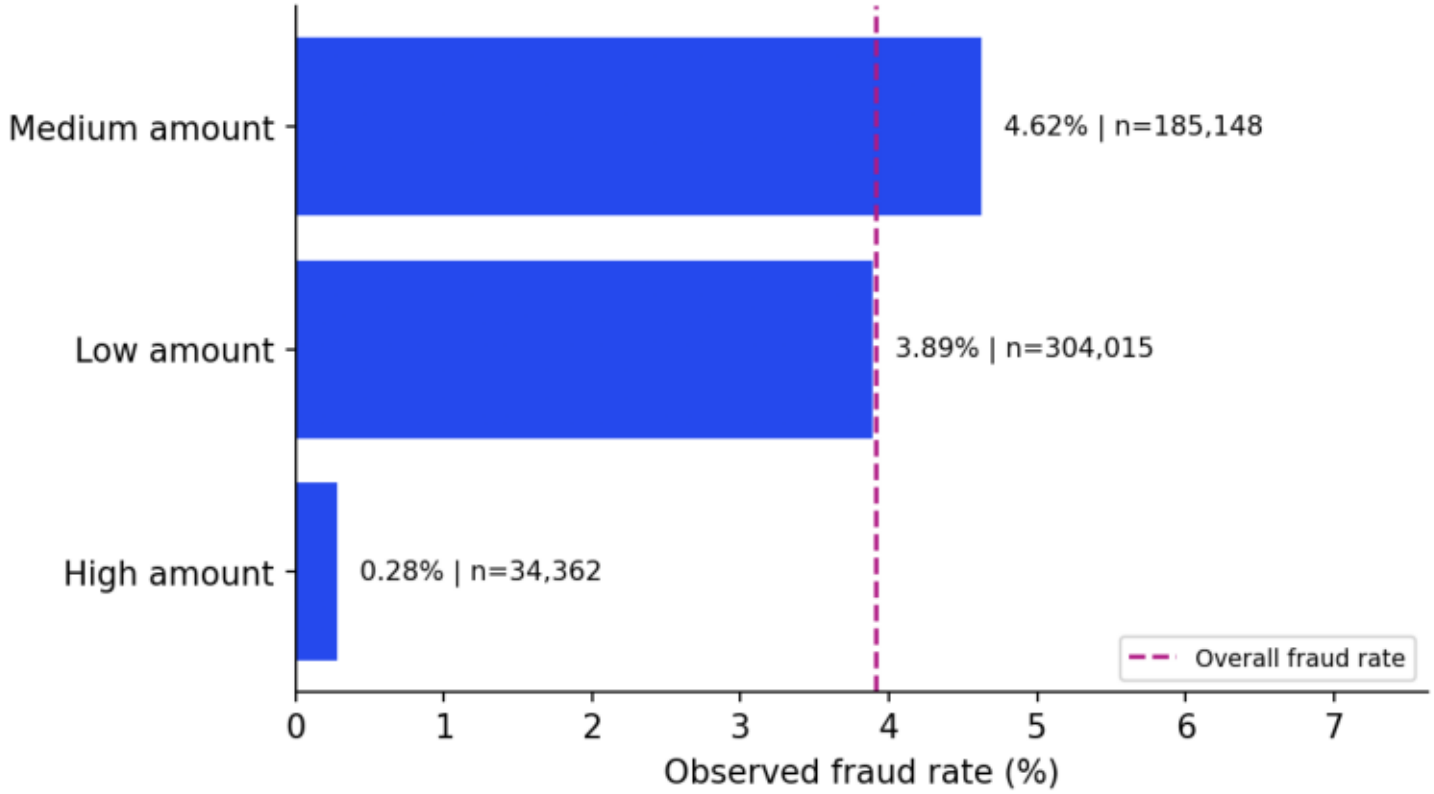
- Fuel: 7.55% observed fraud rate.
- Food & Beverage: 6.63%.
- Evening transactions: 10.24%.
- Mobile: 4.92%; desktop: 2.83%.
- Sao Paulo and Rio de Janeiro: about 5.5%.



Segment rates identify where to investigate, rather than defining automatic blocks.

EDA: verify amount patterns against the data

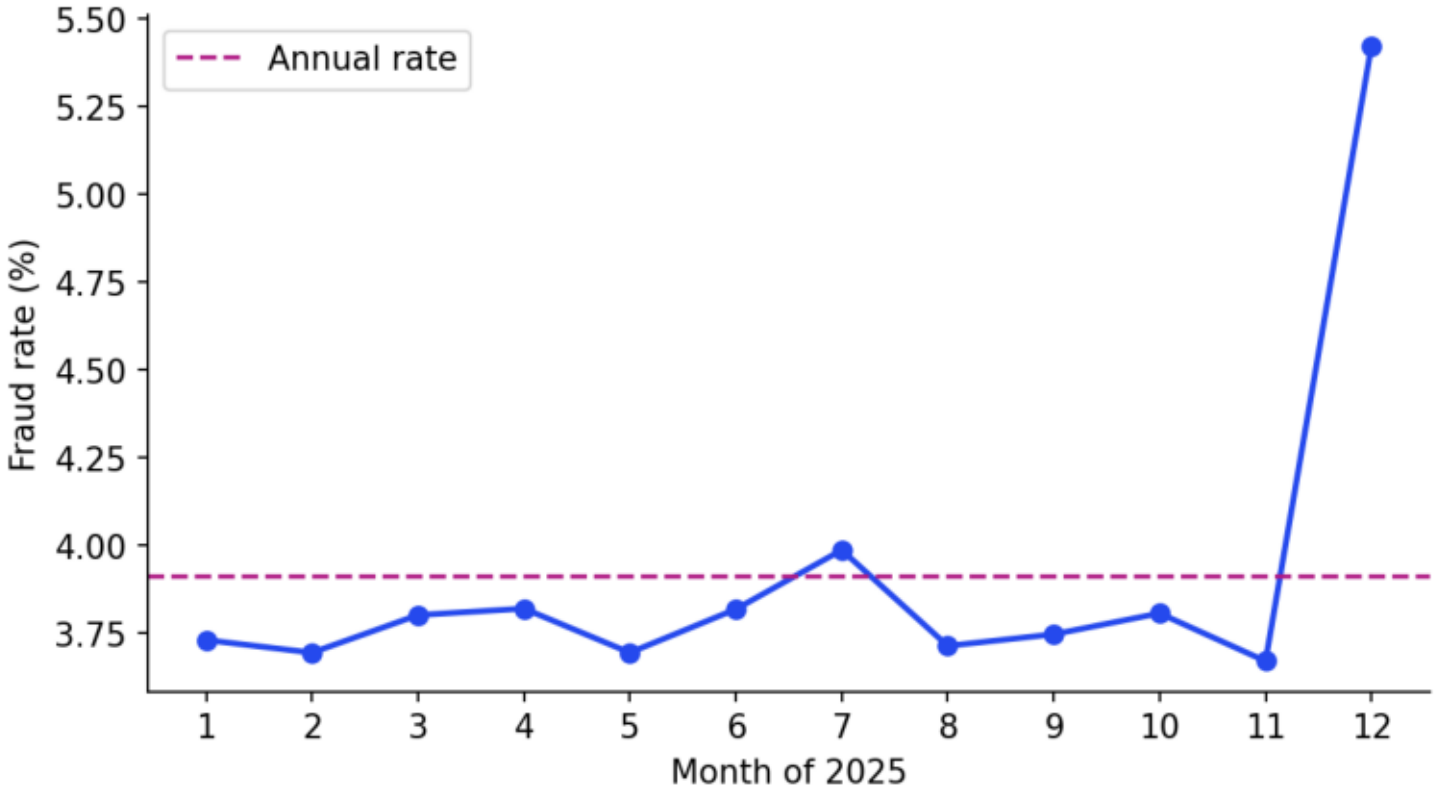
- Medium amount category: 4.62% fraud.
- Low amount category: 3.89%.
- High amount category: 0.28%.
- Median amount: 27.21 for fraud vs 25.11 for non-fraud.
- Higher typical fraud amount does not imply a monotonic category risk.



The latest data do not support treating every high-value purchase as high risk.

EDA: temporal variation across 2025

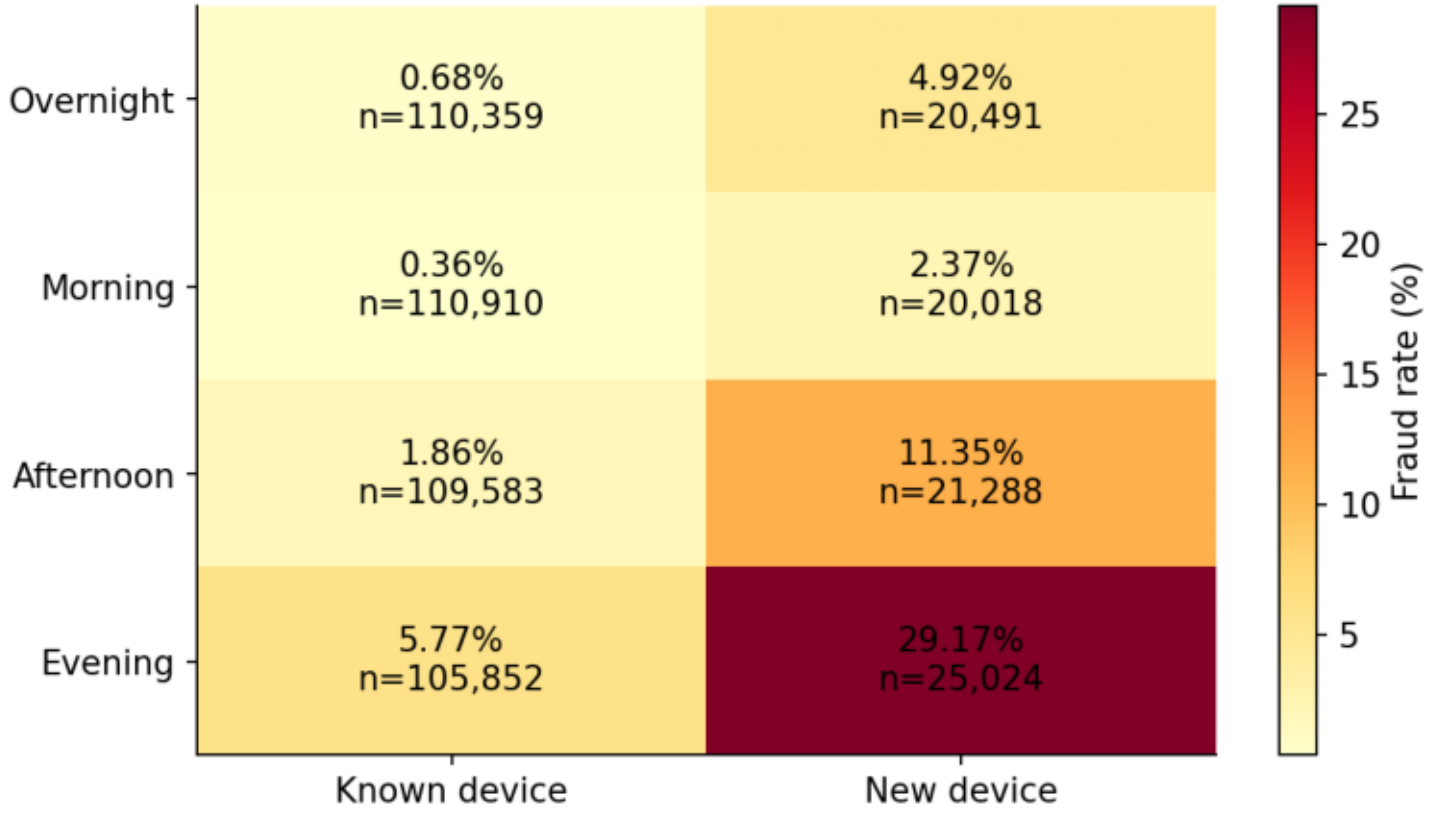
- Observed monthly rates range from 3.67% to 5.42%.
- December fraud rate: 5.42%.
- Month and weekend indicators capture calendar context.
- One year cannot establish recurring seasonality.



Track temporal changes and test performance on future periods.

Multivariate EDA: combine device and time

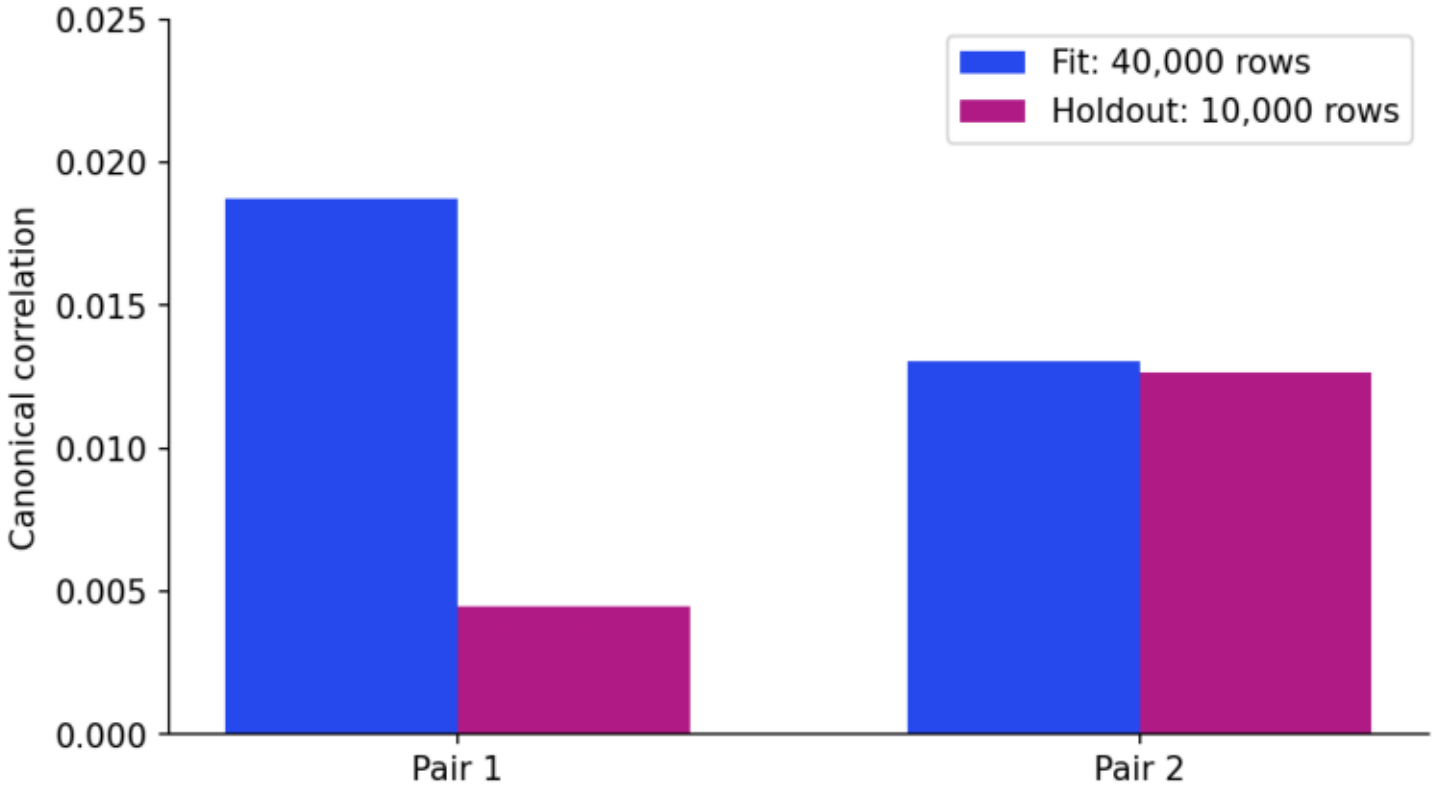
- Compare device novelty within each time period.
- Every cell shows both fraud rate and transaction count.
- The combined view motivates a new-device x evening interaction.
- The notebook also reports descriptive Wilson 95% intervals.



Joint segment analysis adds context beyond separate feature rankings.

CCA: little shared linear structure between blocks

- Numerical block: amount and internal score.
- Context block: cyclical time encodings and device novelty.
- Two pairs of weighted combinations; temperature and target excluded.
- Held-out correlations: 0.0045 and 0.0126.
- Small cross-block association does not imply low predictive value for fraud.



Use this as an exploratory diagnostic, not a reason to discard features.

Feature engineering: customer behavior over time

- Calendar: month, weekday and weekend.
- Purchase counts over the previous 24 hours, 7 days and 30 days.
- Hours since the previous purchase; missing-history indicator.
- Previous 30-day amount mean, sample standard deviation, difference and ratio.
- Changes in city, merchant type, device, time period and amount category.

Behavioral windows use strictly earlier timestamps and preserve missing history.

Interactions, missing history and label availability

- Interactions: new device x evening, city change and relative amount.
- No prior purchases: keep undefined historical statistics as missing.
- Zero historical mean: keep the amount ratio undefined.
- Historical fraud count: previous 30-day labels retain availability assumptions.
- January has no confirmed fraud labels; December supplies recent context.

Label availability must be validated before using historical-fraud features in production.

Training and preprocessing are reproducible

- Random split: 439,761 training / 41,882 validation / 41,882 test rows.
- Fit scaler, encoder and median imputer on training rows only.
- 44 selected features become 50 model inputs after missing-value indicators.
- Save fitted preprocessing and model artifacts; reuse transform() for inference.
- Logistic regression, HGB and Random Forest use balancing; MLP is unweighted.

A random row split does not demonstrate future-period or unseen-customer performance.

Model comparison: validation ranking quality

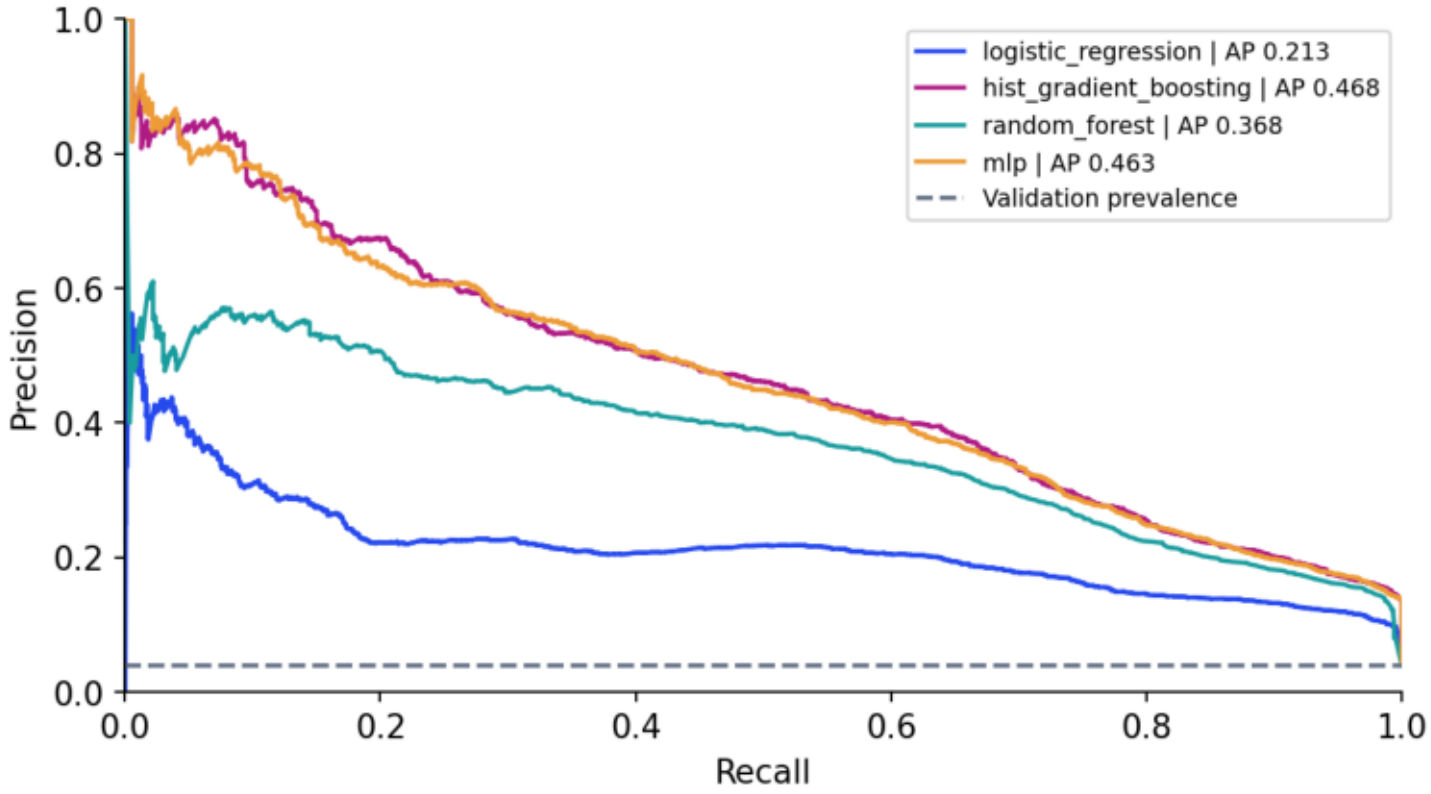
Model	Average precision	ROC-AUC	Recall @ 0.5
logistic regression	0.2132	0.8952	83.8%
hist gradient boosting	0.4679	0.9501	94.1%
random forest	0.3681	0.9381	44.6%
mlp	0.4634	0.9496	20.2%

- HGB achieves the highest average precision: 0.4679.
- MLP is close at 0.4634; operational selection uses the review budget.
- Threshold 0.5 is a descriptive reference; scores are not calibrated probabilities.

Prefer average precision and budget-based recovery for this imbalanced use case.

Precision-recall curves: assess the complete ranking

- HGB and MLP rank fraud similarly across much of the recall range.
- Average precision summarizes ranking across score cutoffs.
- The dashed line is the validation fraud prevalence.
- The budget determines the operating point used for model selection.



A high ROC-AUC is useful context; precision-recall shows the review trade-off more directly.

Threshold construction: turn budget into capacity

- Each reviewed alert costs X, whether fraud or a false positive.
- A batch budget Y allows K = floor(Y / X) reviews.
- Sort scores descending and include the largest affordable full score groups.
- Review when score >= threshold; tied scores receive the same decision.
- If a tied group does not fit, exclude it; unused budget can remain.
- Recompute the threshold for each batch using scores only, without its labels.

Example: X = R4 *and* Y = R10,000 permit at most 2,500 reviews.

Validation: select the model under the same budget

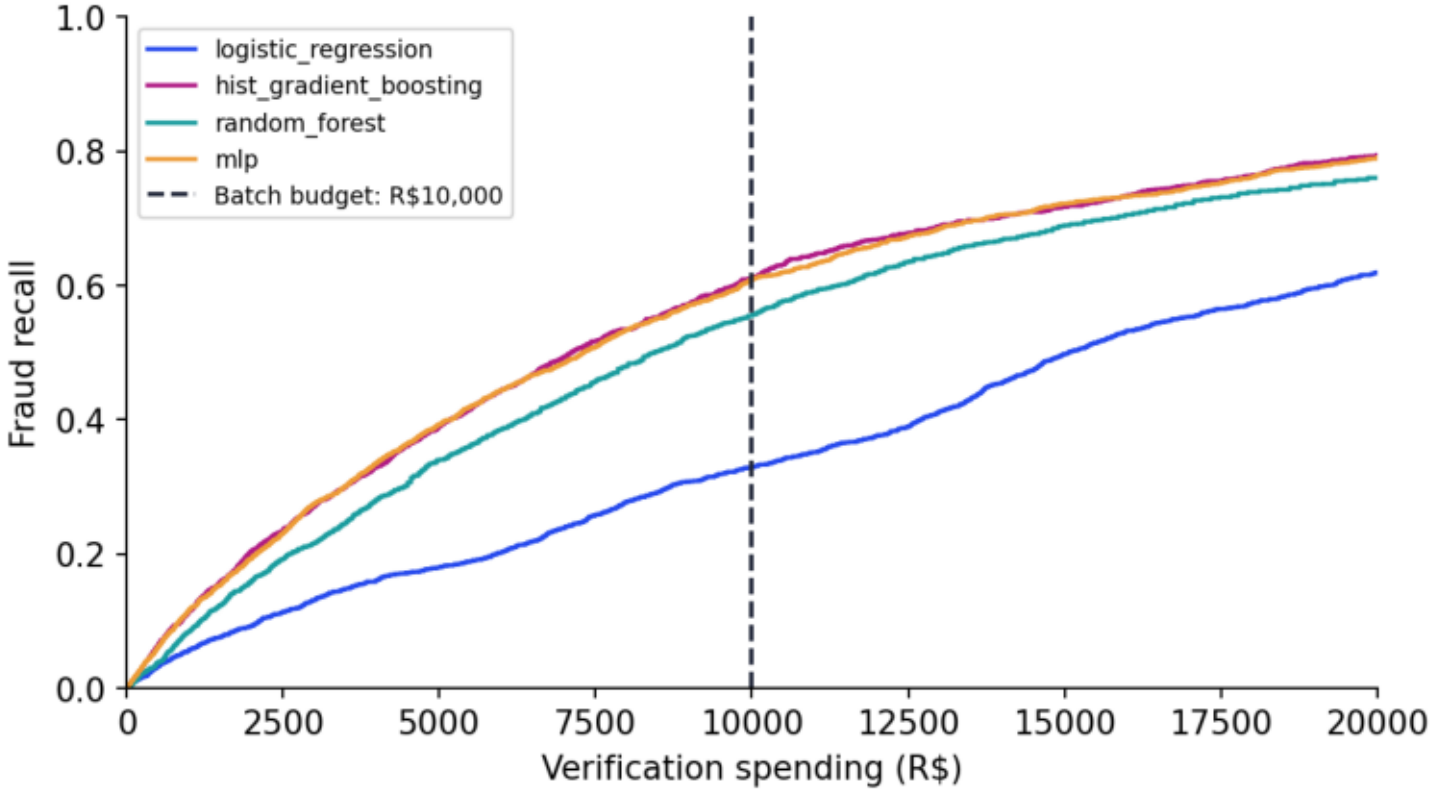
Model	Reviews	Fraud recovered	Precision	Recall
hist gradient boosting	2500	1001	40.04%	61.07%
mlp	2500	998	39.92%	60.89%
random forest	2485	909	36.58%	55.46%
logistic regression	2500	539	21.56%	32.89%

- All candidates use R4*perreviewandaR*10,000 batch budget.
- HGB: 1,001 fraud cases recovered; MLP: 998.
- Select by recovered fraud, then average precision and model name.
- HGB threshold: 0.855291; spending: R$10,000.

HGB wins by a small observed margin; superiority is not established statistically.

Capacity curves show the operational trade-off

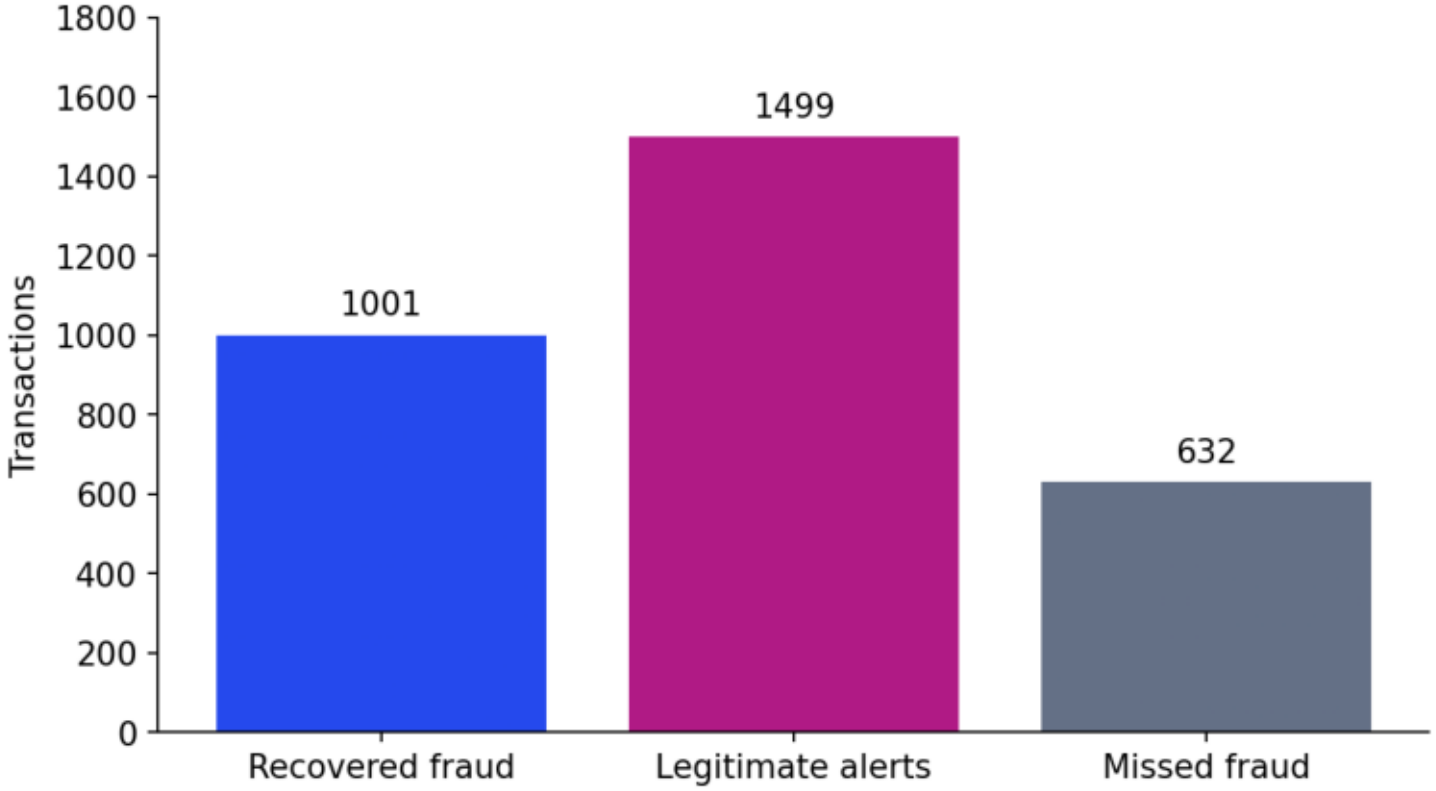
- More review capacity generally recovers more fraud.
- Model ordering can vary with the available budget.
- The current threshold uses only the batch score distribution.
- This policy respects capacity; it does not optimize total financial loss.
- Missed-fraud and incorrect-blocking costs were not supplied.



The threshold follows the review budget rather than an arbitrary fixed score.

Held-out test: recovery under the review budget

- 41,882 test transactions; 1,633 actual fraud cases.
- Model chosen on validation, before test-label evaluation.
- Batch threshold: 0.853015; 2,500 reviews.
- Precision: 40.04%; recall: 61.30%.
- 1,001 recovered fraud cases; 1,499 legitimate alerts; 632 missed fraud cases.



Test spending is R$10,000; threshold recalculation uses scores, not test labels.

January 2026: delivered classifications

- 44,451 transactions, with original columns and row order preserved.
- Budget: R4, 000*forthecompleteJanuarybatch*; *R*4 per review.
- Capacity: 1,000 reviews, or 2.25% of the batch.
- HGB batch threshold: 0.933746.
- Final column flag_fraude: 1 for selected review alerts, 0 otherwise.
- File: datasets/transacoes_jan2026_previsoes.csv

January flags are model decisions, not confirmed fraud; actual precision and recall are unknown.

Conclusions and next steps

- Device novelty and segment combinations motivate behavioral features.
- HGB provides the best observed validation recovery within the configured budget.
- Held-out test recall: 61.30%, with 40.04% precision at 2,500 reviews.
- Deliver 1,000 January alerts within the R$4,000 review budget.
- Next: temporal validation, label-availability audit and feature ablations.
- Monitor drift, realized review precision and budget use; calibrate scores if needed.

The delivery is a reproducible, capacity-aware review prioritization workflow.

Evidence and reproducibility

- EDA: datasets/transacoes_historico_2025.csv and challenge_artefact.ipynb.
- Training: training_workflow.py; preprocessing.py.
- Model metrics: artifacts/training/validation_metrics.csv.
- Budget selection: validation_budget_metrics.csv; budget_threshold.py.
- Test evaluation: test_predictions.csv; labels used only for measurement.
- Delivery: datasets/transacoes_jan2026_previsoes.csv.
- Original presentation preserved; figures and source accompany this update.

All reported operational metrics come from persisted predictions; no models were retrained.